



Original Article

**THE USE OF MECHANISTIC POPULATION MODELS IN METAL RISK ASSESSMENT:
COMBINED EFFECTS OF COPPER AND FOOD SOURCE ON *LYMNAEA STAGNALIS*
POPULATIONS**

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Environ Toxicol Chem., **Accepted Article** • DOI: 10.1002/etc.4391

Accepted Article

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**THE USE OF MECHANISTIC POPULATION MODELS IN METAL RISK
ASSESSMENT: COMBINED EFFECTS OF COPPER AND FOOD SOURCE ON
LYMNAEA STAGNALIS POPULATIONS**

Effects of Cu and food on Lymnaea populations with DEB-IBM

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This article contains online-only Supplemental Data

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Submitted 5 June 2018; Returned for Revisions 1 February 2019; Accepted 6 February 2019

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ABSTRACT

Environmental risk assessment (ERA) of chemicals aims to protect populations, communities and ecosystems. Population models are considered more frequent in ERA as they can bridge the gap between the individual and the population level. *Lymnaea stagnalis* (the great pond snail) is a particularly sensitive organism to various metals, including copper (Cu). In addition, the sensitivity of this species to Cu differs between food sources. The first goal of this study was to investigate if we could explain the variability in sensitivity between food sources (lettuce and fish flakes) at the individual level with a Dynamic Energy Budget (DEB) model.

By adapting an existing DEB model and calibrating it with copper toxicity data, thereby combining information from three studies and two endpoints (growth and reproduction), we put forward inhibition of energy assimilation as the most plausible physiological mode of action (PMoA) of copper. Furthermore, the variation of copper sensitivity between both food sources was considerably lower at PMoA level than at individual level. Higher copper sensitivity at individual level under conditions of lower food quality or availability appears to emerge from first DEB principles when inhibition of assimilation is the PMoA. This supports the idea that DEB explained copper sensitivity variation between food source.

Our second goal was to investigate if this food source effect propagated to the population level. By incorporating DEB in an Individual-Based Model (IBM), population level effects were predicted. Based on our simulations, the food source effect was still present at the population level, albeit less prominently.

Finally, we compared predicted population-level ECx values to individual-level ECx values for different studies. Using the DEB-IBM, the range of effect concentrations decreased significantly: at the individual-level, the difference in chronic EC10 values between studies was a factor 70 (1.13 to 78 µg dissolved Cu/L), whereas at the population level this was a factor

15 (2.9 to 44.6 µg dissolved Cu/L). To improve inter-study comparability, a bioavailability correction for differences in water chemistry was performed with a biotic ligand model. This further decreased the variation, down to a factor 7.4. Applying the population model in combination with a bio-availability correction thus significantly decreased the variability of chronic effect concentrations of Cu for *L. stagnalis*. Overall, the results of the present study illustrate the potential usefulness of transitioning to a more modelling-based environmental risk assessment. This article is protected by copyright. All rights reserved

Keywords

Copper (Metals)

Lymnaea stagnalis (Mollusk)

Dynamic Energy Budget (DEB)

Individual Based Model (IBM)

Population dynamics

Metal risk assessment

Ecological modelling

Population model

1 INTRODUCTION

Conventional environmental risk assessment (ERA) of chemicals is mostly based on toxicity data obtained at the individual (apical) level (e.g. survival, growth, reproduction), therefore not taking into account effects at the population level (EC 2006; ECHA 2008a, 2008b). Laboratory experiments are conducted on various organisms to assess the (long-term) apical effects of metals at the individual level. For data-rich metals, such as Cu, species sensitivity distributions (SSD) are further used to derive the environmental safe threshold concentration for metals in freshwater (ECHA 2008b). The environmental safe threshold concentration or predicted no effect concentration (PNEC) is normally determined as the 5th percentile of the ranked SSD (i.e. the hazardous concentration for 5% of the species) (ECHA 2008b). An issue with SSDs is that they are generally based on chronic effects that occur at the individual level while the goal is to protect populations and communities. In addition, SSDs are generally derived from a variety of toxicity endpoints, such as growth and reproduction, and usually only keep the most sensitive endpoint per species to achieve a conservative estimation (Forbes et al. 2011). Using a common endpoint for all species at the population level (e.g. population growth rate or density) that integrates apical endpoints (even those not considered in the SSD) could improve the relevancy of the SSD approach. Such population-level endpoints generally integrate several individual-level effects (on survival, growth and reproduction) in a single parameter. Furthermore, regulations in ERA are designed to protect populations and the communities they constitute from the effects of harmful substances (Preuss et al. 2009; Galic et al. 2010; Jager et al. 2014). In addition to the above, traditional (individual-level) toxicity endpoints may have a different effect on population level (Galic et al. 2012). Some less sensitive endpoints on individual level could have strong impacts on animal populations (e.g. Miller et al. 2007). Along the same lines, strong observed individual-level effects perhaps make no difference at the population or

ecosystem level (e.g. Martin et al. 2013a). Ultimately, population-level effects integrate the cumulative effects of chemical stressors on individuals, intra-species interactions, and other biotic and abiotic factors (Kramer et al. 2011). Population-level modelling thus decreases uncertainty associated with standard toxicity assessment by predicting the demographic consequences of the effects of chemicals on individuals (Beaudouin and Péry 2013).

For these reasons, modelling tools are increasingly being used as valuable alternatives to conventional techniques in ecological risk assessment (Galic et al. 2010). Population models such as DEB-IBM, which is an individual-based model (IBM) implementation of the dynamic energy budget (DEB) theory, can be used to predict long-term effects of chemicals and environmental stressors on population dynamics (Martin et al. 2013a). DEB and IBM work together in an intuitive mathematical framework (Jager et al. 2014). In such framework, IBM predicts population dynamics through individual behaviour, whereas DEB describes physiological processes on an individual level. Together, they can link individual metabolism to population dynamics. Moreover, DEB-IBM offers a mechanistic framework for implementing the impact of stressors on animal metabolism (i.e. energy assimilation, structural growth, maturation, structural and maturity maintenance, and reproduction) which can then be extrapolated to the population level (Martin et al. 2012; Jager et al. 2013).

Lymnaea stagnalis (the great pond snail) is a particularly sensitive organism to various metals, such as Pb, Ni, Co, Ag and Cu (Grosell and Brix 2006; De Schamphelaere et al. 2008; Brix et al. 2011; Ng et al. 2012; Berteloot et al. 2015; Côte et al. 2015; Crémazy et al. 2018). An extensive amount of literature on the toxic effects of Cu on early life stages of *L. stagnalis* is available. Additionally, studies on the effects of Cu on the growth and reproduction of the closely related species *Lymnaea luteola* (the apple snail) are available. Unpublished data

suggests that the sensitivity of *L. stagnalis* largely depends on the available food source (e.g. lettuce vs. fish flakes) (Berteloot et al. 2015). Elsewhere, it has been shown that the quality of food can alter the growth pattern of individual snails, suggesting the energy budget of *L. stagnalis* depends on the food quality (Zimmer et al. 2012, 2014). Since it has been shown that the food source influences sensitivity of *L. stagnalis* individuals to Cu, differences in population-level effects could also be observed for populations that have access to different food sources. The variability in effect levels of copper between different studies is very high (up to a 70-fold variation) (Brix et al. 2011; Berteloot et al. 2015; Crémazy et al. 2018). Several factors have been reported to contribute to this variability, including genetic diversity, the food source, and differences in water chemistry (Berteloot et al. 2015; Côte et al. 2015). Additionally, the potential roles of the calcium metabolism and the potentially different feeding regimes between juvenile and adult snails have been discussed in this context (Brix et al. 2011; Das and Khangarot 2011). While the relevant OECD 243 testing guideline covers testing adult individuals (OECD 2016), most studies have been performed with juvenile snails. Although the toxicity of metals to *L. stagnalis* has been thoroughly investigated (Grosell and Brix 2009; Brix et al. 2011; Brix et al. 2012; Crémazy et al. 2018), the mechanisms of toxicity (including *molecular initiating events* and *key events* at higher levels of biological organisation, as defined in the adverse outcome pathways, AOP; Ankley et al. 2010) as well as factors influencing metal toxicity for this species remain insufficiently understood.

Our main goals were to investigate (i) if we could explain the variability in sensitivity at the individual (apical) level of *L. stagnalis* to Cu between food sources with a DEB model, (ii) whether this variability was still present when extrapolating the effects to the population level, and (iii) how (extrapolated) effect concentrations (ECx) at the population level compared with ECx values reported at the individual level. To achieve these goals, we first addressed

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differences in the energy budget caused by the food source alone, which is an important factor. To this end, a general assimilation correction factor was implemented to account for food source effects on energy assimilation. Secondly, data on chronic Cu toxicity (i.e. Cu effects on *L. stagnalis*) were integrated in a DEB-IBM to account for toxic effects of Cu. The data from Berteloot et al. (2015) did not give a decisive answer as to which DEB process was the most plausible one to be affected by Cu. Therefore, data on a closely related species (*L. luteola* – Das and Khangarot 2011) were used as well, since simultaneous effects on growth and reproduction were available from that study, which allowed us to put forward the most plausible DEB-level physiological mode of action (PMoA). Third, with the calibrated DEB-IBM, extrapolated population-level effects of Cu to *L. stagnalis* population were predicted (for Berteloot et al. 2015) to examine if the food source effect propagated to the population level, and to compare ECx values between studies. For the latter, a bioavailability correction (due to differences in water chemistry) was performed using a Biotic Ligand Model (BLM). Finally, based on the performed simulations, ECx values on population-level endpoints (e.g. equilibrium population density) were derived and compared to their traditional (individual-level) counterparts. The study of Brix et al. (2011) was therefore extrapolated as well, as they reported a very different ECx value for *L. stagnalis*.

2 MATERIAL AND METHODS

2.1. *Reported Cu toxicity data*

Cu toxicity data of *Lymnaea* spp. from literature and unpublished research was compiled (Table 1). Data is available on Cu effects on mortality and growth of *L. stagnalis* (i.e. acute toxicity data from Ng et al. (2012) and Côte et al. (2015); chronic toxicity data from Brix et al. (2011) and Berteloot et al. (2015)). Additionally, data on embryonic development, growth and reproduction of the related snail species *L. luteola* are available (Das and Khangarot 2011; Das et al. 2012). The raw data from Berteloot et al. (2015) was available (the raw data is available on Figshare, see Supplemental Information). Their study reported the chronic effects of Cu on survival and growth of *L. stagnalis* in combination with two different food sources (lettuce and fish flakes). In addition, for effect calibration, the data of Brix et al. (2011) was used as well, as they reported the lowest ECx value for *L. stagnalis* (and thus the most sensitive response) at the apical level. Das and Khangarot (2011) simultaneously investigated effects on both reproduction and growth of *L. luteola*, whereas Berteloot et al. (2015) only investigated effects on growth. Since no reproduction data for *L. stagnalis* was available, we used the reproduction patterns observed in *L. luteola* to calibrate our DEB model. These species are closely related (same genus), so we assumed the toxicity patterns are comparable. To derive the physiological mode of action (PMoA) of Cu toxicity on *Lymnaea* spp. metabolism, the data of Das and Khangarot (2011) was used as well.

2.2. *Modelling principles*

A generic individual-based model (IBM) implementation of the dynamic energy budget (DEB) theory was developed by Martin et al. (2012). In short, IBMs are models in which each individual in the population is simulated as a single entity. Population dynamics emerge from the processes modelled at the individual level. In DEB-IBM, the DEB theory is used to describe

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physiological process occurring at the individual level. The individual-level (apical) effects emerge from these DEB processes. Population dynamics emerge from the DEB and IBM equations that are imposed. A full description of the original model of Martin et al. (2012) as well as the implementation file is accessible on the web (<https://popecology.wordpress.com/deb-ibm/>).

We adapted the original model of Martin et al. (2012) to account for the life history of *L. stagnalis*. Originally, the DEB-IBM of Martin et al. (2012) was developed to simulate *Daphnia magna* populations (Martin et al. 2013a, 2013b). Zimmer et al. (2012) calibrated a full DEB model for *L. stagnalis*. The DEB parameters for *L. stagnalis* calibrated by Zimmer et al. (2012) were used in our *L. stagnalis* DEB-IBM in the present study (Table 2). In addition, Zimmer et al. (2014) observed that the growth curve of *L. stagnalis* deviates from the Bertalanffy growth curve (von Bertalanffy, 1934) and developed a metabolic acceleration sub-model that corrects for the accelerated growth in the snails. We included the metabolic acceleration sub-model proposed by Zimmer et al. (2014) in our *L. stagnalis* DEB-IBM as well.

Our DEB-IBM included two more adaptations. First, we introduced a general assimilation correction factor to account for differences in the dynamic energy budget of non-Cu-exposed *L. stagnalis* between different experiments. This assimilation correction factor is analogous to the food quality factor of Zimmer et al. (2012), which accounts for differences in food quality (e.g. lipid content, caloric value, palatability, etc.) and food availability when an identical genetic strain is used. But, in contrast to Zimmer et al. (2014), the assimilation correction factor can also account for differences between studies with different genetic strains (e.g. the European strain from Berteloot et al. 2015, and the American strain from Brix et al. 2011) or with different experimental set-up. The assimilation correction factor as defined and as applied

here is thus more broadly applicable than the food quality factor from Zimmer et al. (2014). Second, we implemented lethal and sub-lethal effects of Cu as sub-models. These adaptations are discussed in more detail in Sections 2.2.1, 2.2.2 and 2.2.3. The development of our *Lymnaea* DEB-IBM was done in the software platform NetLogo (Wilensky, 1999). A pseudo-code of our model with the specific adaptations can be found in Section 5 of the Supplemental Information. In addition, the implementation of our *Lymnaea* DEB-IBM for Cu toxicity in NetLogo can be downloaded from Figshare (see Supplemental Information).

2.2.1. Food source effect

Berteloot et al. (2015) investigated the effect of food source (i.e. lettuce vs. fish flakes) and Cu on survival and growth of *L. stagnalis* (after 28 days). Their results show that individual growth was much larger in snails fed with TetraMin fish flakes (Tetra GmbH, Melle, Germany). The authors reported a mean dry blotted weight of 50.3 mg in the fish flakes fed control, while a mean weight of 4.2 mg was noted in the lettuce fed control after 28 days. To account for the difference in quality between food sources, an extra parameter was implemented, an assimilation correction factor (a_f), of which the value is determined by the food source (as in Zimmer et al. 2012). The energy assimilation rates in our DEB-IBM were therefore corrected for the food source by calibrating the assimilation correction factor (a_f) for the control data (no Cu exposure) for *L. stagnalis* growth of Berteloot et al. (2015).

Mathematically, the assimilation correction factor a_f is incorporated in the equation that determines the assimilation flux, S_A [J/h] (Equation 1). Higher values for a_f will lead to higher assimilation fluxes. In Equation 1 the assimilation flux S_A (i.e. the amount of assimilated energy per unit of time) of an individual depends on the scaled functional response (f), the size of the

snail (L), the maximum surface-specific assimilation rate (p_{Am}), and the assimilation correction factor (a_f):

$$S_A = a_f * f * L^2 * p_{Am} \quad (1)$$

Assimilation correction factors were estimated for lettuce and fish flakes based on the control data of Berteloot et al. (2015). An assimilation correction factor was determined for the data of Brix et al. (2011) as well. Brix et al. (2011) used a mixture of lettuce, sweet potato and carrots for the culture of *Lymnaea*. For the experiment, the authors used a mixture of lettuce and sweet potato, resulting in a control weight of 66.9 mg after 30 days. The assimilation correction factor was determined based on their control growth data (i.e. no Cu exposure). It should be noted that for Berteloot et al. (2015) the assimilation correction factor can be interpreted as a food quality factor, since it concerns snails from the same strain exposed under similar conditions but with different food sources (lettuce and fish flakes). The assimilation correction factor determined for Brix et al. (2011) also includes differences in genetic strain and test design as compared to the Berteloot et al. (2015) study. The calibration procedure for the assimilation correction factor is described in Section 2.3.

2.2.2. Lethal effects

Effects on survival (i.e. increased mortality) due to exposure to Cu were implemented into the model by increasing the mortality rate with the exposure concentration. The mortality rate can be determined for a given Cu concentration (Equation 3; De Laender et al. 2008). For a given toxicity experiment, the mortality rate $m(Cu)$ was determined by the $LC50$ and slope β of the determined dose-response curve, as well as the exposure duration t_{exp} of the experiment.

$$m(Cu) = \frac{1}{t_{exp}} * \ln \left(1 + \left(\frac{Cu}{LC50} \right)^\beta \right) \quad (3)$$

From the mortality rate, the stochastic probability of dying within each time step can be calculated (Equation 4). An individual will survive each time step (in our model, each time step is one day) depending on the stochastic probability of dying $p_m(Cu)$.

$$p_m(Cu) = (1 - e^{-1 * m(Cu)}) \quad (4)$$

The values for $LC50$, β and t_{exp} were used as reported from the studies (Table 1). Berteloot et al. (2015) only investigated effects on juveniles. In absence of data on adults, we assumed the same mortality parameters for adults and juveniles. In addition, there were no statistically significant differences in mortality between the lettuce and fish flakes fed snails. Hence the same mortality parameters were used, i.e. an $LC50$ of 109 $\mu\text{g/L}$, β of 2.1, and t_{exp} of 28 days.

2.2.3. Sub-lethal effects

Sub-lethal effects in DEB theory can be implemented via the so-called Physiological Mode of Action (PMoA). The PMoA is described as the large-scale DEB process on which a stressor can have an effect (Martin et al. 2013a; Ashauer and Jager 2018). The stressor will impact a specific DEB process (determined by the PMoA), changing the energy flows (i.e. metabolism) of the organism. Eventually, this will change their pattern in growth, reproduction, maturation, etc. Five PMoAs are suggested by Martin et al. (2013a): inhibition of energy assimilation of (1), increase in costs of (maturity and structural) maintenance (2), increase in costs of structural growth (3), decrease in reproduction efficiency (4), and decrease in embryonic survival (5). Martin et al. (2013a) used a stress level s which depends on the Cu exposure concentration. The stress level is linked to a specific DEB process or parameter in accordance with the chosen PMoA. Table 3 gives an overview of the affected DEB parameters per PMoA and their mathematical link with the stress level. The most plausible PMoA is determined by calibrating

the model with the different PMoAs and determining the PMoA that best predicts the observed effects from the data.

A log-logistic relationship was selected to determine the stress level s as a function of the Cu exposure concentration (Equation 5). Note that for a log-logistic relationship the value for the stress level s will always be between 0 and 1, with 0 meaning there is no stress and 1 meaning the physiological process is completely disrupted (e.g. no assimilation of energy takes place). The log-logistic relationship is described by two parameters which have to be calibrated, the median effect concentration $EC50_{PMoA}$ and the intrinsic slope β_{PMoA} :

$$s(Cu) = \frac{1}{(1 + \exp(\beta_{PMoA} * (\ln(Cu) - \ln EC50_{PMoA})))} \quad (5)$$

The values of the effect parameters for sub-lethal effects, $EC50_{PMoA}$ and β_{PMoA} (Equation 5), were calibrated based on Cu toxicity data. The calibration procedure of effect parameters is described in Section 2.3.

To determine the PMoA of Cu, the complete dataset on snail growth from Berteloot et al. (2015) was considered. Because Berteloot et al. (2015) investigated effects on juvenile snails (freshly hatched), only effects on growth were reported. No effects on reproduction were reported as the exposure time was 28 days, while it is reported that *L. stagnalis* reaches sexual maturity in the wild at an age between 3 and 4 months (Noland and Carriker, 1946), or after 32 days in the laboratory (Munley et al. 2013). The PMoAs for reproduction efficiency and embryonic survival were thus not considered. Calibration of the different PMoAs was performed on the growth datasets for lettuce and fish flakes separately. This allowed to test the hypothesis that the quality of the food source interferes with the sensitivity of the PMoA of the snails. The respective assimilation correction factors for both diets were considered. Effect parameters

were thus determined for each food source separately and for each PMoA based on the data of Berteloot et al. (2015). Similarly, effect parameters for Brix et al. (2011) were determined based on the dry blotted weight data (converted to length, assuming a weight/length relationship of $W = L^3 * 0.0454 \text{ mg/cm}^3$; based on Berteloot et al. 2015).

Available data was not sufficient to unequivocally determine one PMoA. Therefore, analysis was performed on the effect data from Das and Khangarot (2011). Since they studied effects of Cu on growth and reproduction simultaneously over 49 days, it was expected that a calibration on this dataset would give more insight in the underlying DEB mechanism of toxicity on *Lymnaea* spp. Because of growth and reproduction differences between *L. luteola* and *L. stagnalis* (same genus but different species), the relative effect of Cu on growth and reproduction was used to determine the PMoA. Additionally, since raw data from the control treatment was not available, no food correction (i.e. an assimilation correction factor of 1 – the default value) was considered here.

2.3. Calibration

For calibration of the newly introduced assimilation correction factor (Section 2.2.1) and effect parameters (Section 2.2.3), Approximate Bayesian Computation (ABC) was used (van der Vaart et al. 2016). For each dataset, the model simulations were iterated 10 000 times with a randomly sampled value for the assimilation correction factor per iteration. Based on the Euclidean distance ρ (Equation 2), the 100 best fits were selected, and the mean and standard deviation of the parameters were determined. The Euclidean distance between data and predictions was used to assess the goodness of fit of the model (van der Vaart et al. 2016). The Euclidean distance is determined as:

$$\rho(m_i, D) = \sqrt{\sum_j \left(\frac{m_{i,j} - D_j}{sd(m_j)} \right)^2} \quad (2)$$

where $m_{i,j}$ is the model output of endpoint j in iteration run i , D_j is the observed data of endpoint j , and $sd(m_j)$ the standard deviation of endpoint j of all simulation runs. Data processing as well as the calibration were performed in R-Studio (R Foundation for Statistical Computing, Vienna, Austria). Because the Euclidean distance ρ scales the sum of square errors (SSE) of both endpoints by the variance of each endpoint (Equation 3), this measure can be used to calibrate a model for endpoints with different units (i.e. number of produced eggs vs. snail size in mm).

2.4. Population effects with DEB-IBM

Simulations with the calibrated *Lymnaea* DEB-IBM were performed to predict population-level effects for different food sources (i.e. the conditions of Berteloot et al. 2015, and Brix et al. 2011). The output data of the population simulations were linked to relevant population-level endpoints for risk assessment purposes, e.g. population equilibrium density (number of individuals), population age structure (number of individuals per life stage – embryos, juveniles and adults), and the population reproduction (number of eggs per adult per day). Additionally, the effect of Cu and food source (i.e. lettuce and fish flakes as in Berteloot et al. 2015, and mixed as in Brix et al. 2011) at the population level was assessed.

2.4.1. Population simulations

An overview of all the simulations performed with the model can be found in the Supplemental Information (see Table S1). Simulations for population effects were performed under constant (steady-state) conditions (i.e. constant food level, constant temperature of 20 °C, constant Cu concentrations, and constant water chemistry), and for each set of conditions the model was run

in 100 iterations. The food density was set constant at 200 J/m² (or 8.72 mg dry weight/m² for periphyton – an average food concentration of periphyton in a freshwater pond; Frost et al. 2005). Cu concentrations were varied from 0 to 100 µg/L. To predict population-level effects, the best fitting PMoA as determined for the individual-level toxicity data was used. For both the two food sources of Berteloot et al. (2015) (lettuce and fish flakes) as well as for Brix et al. (2011) population-level effects were determined.

RStudio was used to process model simulations and determine concentration effect relationships (RStudio Team 2016). Using the drc package in R (Ritz et al. 2015), a three-parameter logistic model was fitted to establish a relationship between population level endpoints and the environmental Cu concentration (Equation 6). The parameter d indicates the maximum response (i.e. the response of the control treatment), b is the slope of the curve, and e is the natural logarithm of the median effect concentration (i.e. the EC50):

$$f(Cu) = \frac{d}{(1 + \exp(b * (\ln(Cu) - e)))} \quad (6)$$

2.4.2. Correction for the water chemistry

Brix et al. (2011) have used a lower DOC content in their test set-up compared to Berteloot et al. (2015), which might explain the higher sensitivity to Cu at the individual-level reported by the former authors (Table 1). A correction with a biotic ligand model (BLM) was performed to account for the effect of bioavailability. Normalisation was performed using BLM software (HydroQual 2015) incorporating the Windermere Humic Aqueous Model (WHAM) number V (Tipping 1994) for Cu speciation. In addition, the chronic Cu invertebrate BLM (De Schamphelaere and Janssen 2004) was used to calculate biotic ligand concentrations from ion activities. The BLM translates dissolved Cu concentrations to a biotic ligand concentration, i.e.

a Cu concentration accumulated on the biotic ligand of the organism (in nmol Cu / g-wet BL) (for details, see Supplemental Information, Section 6). The dose-response curves at both the individual (as reported in Table 1), the PMoA level (Equation 5) and population level (Equation 6) were thus corrected for the water chemistry. This allowed to compare Cu toxicity ECx values between different studies (Berteloot et al. 2015, and Brix et al. 2011) and between different levels of organisation (individual, PMoA and population level).

3 RESULTS

3.1. Food-related effects

Snail growth predicted by our DEB-IBM model was calibrated against the control growth data (length) from Berteloot et al. (2015) to account for each food source. Assimilation correction factors of 1.3 for lettuce (Figure 1a) and 4.0 for fish flakes (Figure 1b) were obtained. The estimated value for the assimilation correction factor for Brix et al. (2011) is 3.8, which is comparable to the value obtained for the fish flakes diet, but higher than for the lettuce-only diet in Berteloot et al. (2015). Note that the assimilation correction factor for Brix et al. (2011) also corrects the DEB model for differences in genetic strain (since an American strain of *L. stagnalis* was used by Brix et al. 2011). The default value for the assimilation correction factor is 1, which means the DEB parameters as calibrated by Zimmer et al. (2014) are used. The default values account for the genetic strain and food source conditions used by Zimmer et al. (2014) to calibrate the DEB parameters for *L. stagnalis*.

3.2. Physiological mode of action

Since lethal effects for *L. stagnalis* occur at much higher Cu concentrations than sub-lethal effects, we will focus here on the PMoA determined. Based on the Euclidean distance between data and the model, the PMoA for decrease in energy assimilation best fits (i.e. lowest value for ρ) the decreasing trend in growth with increasing concentrations observed by Berteloot et al. (2015) (Table 4). Differences with the other PMoAs are, however, too limited (see Supportive Information: Figures S1-S6) to decide which is biologically the most plausible PMoA (Table 4). Since there is no data on reproduction, it could be either one of the three that is related to Cu toxicity effects on *L. stagnalis*.

Therefore, the data for a closely related species were used (*L. luteola*). Based on this additional data, the PMoA for decrease in energy assimilation was the only one to simultaneously capture the decreasing trends in both growth and reproduction (Figure 2a). For the PMoAs for increase in cost of growth (Figure 2b) and increase in cost of maintenance (Figure 2c) there is a clear mismatch for the effect on reproduction. The PMoA for decrease in assimilation also gave statistically the best fit compared to the other PMoAs (see Supplemental Information; Table S3). Altogether, the PMoA of reduced energy assimilation is the most plausible PMoA among those considered. Figure 3 shows the predictions of the model with the PMoA for inhibition of assimilation for the data of Berteloot et al. (2015) (see Supplemental Information, Figure S7 for Brix et al. 2011).

3.3. Food source effect at the population level

Population simulations were performed with our DEB-IBM calibrated for the PMoA of decrease in energy assimilation for the data of Berteloot et al. (2015). Population biomass over time is shown for the two food sources: lettuce (Figure 4a) and fish flakes (Figure 4b) (food and effect parameters based on Berteloot et al., 2015) and two Cu exposure concentrations (0 and 30 µg dissolved Cu/L). The population initially grows to a peak density after which an equilibrium is reached. Compared to the control, the 30 µg dissolved Cu/L simulation shows a lower equilibrium population density. The decrease is smaller for fish flakes than for lettuce.

Effect concentrations on equilibrium population density and other population-level endpoints were determined based on the population simulations (Table 5). The predicted EC10 (µg dissolved Cu/L) values for the different population endpoints to *L. stagnalis* fed lettuce are similar: approximately 15 µg/L. For fish flakes predicted EC10 values are approximately 50 µg/L, indicating this snail population is less sensitive to Cu when fed with fish flakes. The

difference between lettuce and fish flakes decreased when extrapolating to the population level, from an 11-fold at the individual-level (endpoint: dry bottled weight after 28 days), to a 2.3-fold difference at the population level.

3.4. Comparison between studies

In order to compare between studies, a correction with a biotic ligand model (BLM) was performed to consider the effect of bioavailability. Since Berteloot et al. (2015) and Brix et al. (2011) used different test media, such a BLM correction is necessary to compare the effects between both studies at the individual and population level. In Table 6, the corrected EC10 values at the individual level (expressed as BL-Cu concentrations) show less variation (53-fold) than when uncorrected (expressed as dissolved concentration; 70-fold). This suggests that the low DOC concentration used by Brix et al. (2011) contributed somewhat to the higher effects of copper on the snails in their experiment (since there is more metal bioavailable). At the population level, the variation in EC10 values is further reduced, with the minimum and maximum EC10 values expressed as BL-concentrations being a factor 7.4 apart.

Figure 5 summarises the extrapolation of Cu toxicity effects from the individual (apical) level (Figure 5a) to the PMoA level (Figure 5b), and ultimately to the population level (Figure 5c). The left figures show the dose-response curves based on dissolved Cu concentrations, whereas the right side shows the curves corrected with the BLM (in biotic ligand Cu concentrations). Looking at the non-bioavailability-corrected curves, the range in EC10 values (vertical dashed lines) is reduced moving from the individual to the PMoA and to the population level. The variation in EC10 values between the studies is thus reduced when extrapolating to the population level. Using a BLM correction, the range in EC10 values at the population is further reduced.

4 DISCUSSION

4.1. Importance of food quality for *Lymnaea* populations

For *Daphnia magna*, it has been shown that a low nutritional value impacts ingestion, assimilation and reproduction rates, and therefore reduces their overall fitness (Arnold 1971; Bøhn et al. 2008). However, this does not explain the interaction between the food source and sensitivity to a stressor found in the current study. Studies on the interactive effects between food and metal sensitivity are limited. Naddy et al. (2011) investigated toxicity of freshwater organisms to Ag with and without food. The results of exposure of four different species to Ag show an increase in acute EC50/LC50, ranging from a factor 1.25 to almost 22.4 in presence of food. It could be argued that the absence of food might induce starvation in the exposed animals, potentially increasing their overall sensitivity to stress. Regardless, Naddy et al. (2011) also observed that the food source affects acute Ag toxicity in invertebrate species. Up to a twofold difference in acute EC50 was found for *D. magna* fed with two different food sources (a 3:5 and 1:5 mixture of yeast, wheat grass and trout chow (YCT) (1), and algae (2) – *Raphidocelis subcapitata*, formerly known as *Pseudokirchneriella subcapitata*). Besser et al. (2016) also found differences in lead (Pb) toxicity to *Hyaella azteca* between two different diets. No significant differences in *H. azteca* survival were observed between an YCT diet and algae/fish flakes diet. In contrast, significant effects on fecundity and weight were observed by the authors in the YCT diet, which were not observed with the algae/fish flakes diet. The results from Berteloot et al. (2015), Naddy et al. (2011) and Besser et al. (2016) highlight the importance of the food source on the sensitivity of invertebrates to metal toxicity.

The data from Berteloot et al. (2015) indicates that growth strongly depends on the food source, and the low protein content of lettuce compared to fish flakes could lead to lower energy assimilation rates in the snails (Zimmer et al. 2012; Berteloot et al. 2015). In the present study, This article is protected by copyright. All rights reserved

an assimilation correction factor of 1.3 and 4.0 was found for lettuce and fish flakes respectively. In comparison, Zimmer et al. (2012) found a food quality factor of 1.2 and 1.6 for lettuce, and 2.4 for fish flakes. A higher value for the assimilation correction factor for fish flakes is generally found, indicating that fish flakes are more suitable as food for juvenile *L. stagnalis*. Moreover, from the Brix et al. (2011) study a higher assimilation correction factor of 3.8 could be calculated. Brix et al. (2011) used a mixture of lettuce and sweet potato as food source. The exact ingredient composition (in percentage) is unknown so it is impossible to determine what causes the higher assimilation correction factor. The snails could have taken advantage of biofilm growth on sweet potato explaining the higher assimilation correction factor, but possibly the higher assimilation correction factor is due to a strain effect (i.e. caused by a difference in genetic strain).

Lettuce and fish flakes thus impacted the expected growth curve of *L. stagnalis*, as well as their sensitivity towards Cu. However, it is still unclear how a natural diet affects the growth of snails. In addition, life stage related diet composition and its effect on the metabolism for the snails remains largely unknown. Observations indicate that *L. stagnalis* grows to much larger sizes in the wild than under laboratory conditions (Berrie 1966; Zimmer et al. 2014). Moreover, the effect of natural food on the sensitivity to Cu is also unknown. The simulations suggest that food of good quality (e.g. high protein content food such as fish flakes) decreases population sensitivity to Cu.

Elger and Lemoine (2005) investigated the palatability of various macrophytes to *L. stagnalis*. They concluded that *L. stagnalis* prefers high protein content food with low dry matter content (DMC). However, next to these (bio)chemical traits, structural characteristics of the food are also important (Elger and Lemoine 2005; Zimmer et al. 2012). Certain defence mechanisms of

macrophytes (e.g. tissue toughness) against herbivores may hinder *L. stagnalis* in their feeding (Elger and Lemoine 2005). The structural softness of the food source, which determines its accessibility, is an important factor as well. The snails seem to prefer softer food, which might explain their affinity for fish flakes over lettuce (Elger and Lemoine 2005; Berteloot et al. 2015). It is suspected that *L. stagnalis* feeds primarily on periphyton and biofilms during juvenile life stages, and on detritus and macrophytes during later life stages (Zimmer et al. 2012). Because the diet between juveniles and adults differs, different assimilation correction factors depending on the life stage might be applicable. Table 7 gives an overview of the various (potential) food sources for *L. stagnalis* with their DMC and protein content range. Fish flakes have the highest protein content (i.e. 43.2% of wet weight), and the protein content of lettuce is noticeably lower (1 and 4.8% on wet weight basis) (McKeehen et al. 1996). In the present study an analysis was performed on the protein content of green, organic lettuce (Table 7). The lower protein content of organic lettuce compared to fish flakes might explain the lower value of the assimilation correction factor. Assimilation correction factors were determined for only two sources (excluding Brix et al. 2011), and therefore no meaningful quantitative relationship between the assimilation correction factor and the DMC or protein content of the food source could be established.

4.2. Physiological mode of action for Cu toxicity in *Lymnaea*

Our analysis showed that access to multiple endpoints and independent datasets is essential to put forward a more plausible hypothesis as to the PMoA of a chemical stressor than what would be possible from a single endpoint and/or a single dataset. It provides the modeller additional data for calibration and validation. We therefore argue that it is crucial to gather and analyse growth and reproduction data simultaneously, since it allows the PMoA of the chemical stressor to be identified more plausibly.

Our finding that the PMoA of inhibition of energy assimilation explains the data best, leads to the hypothesis that Cu influences certain (sub-)organism level processes (e.g. cellular) that result in a decrease in the build-up of energy reserves, and ultimately in a decrease of snail growth and reproduction. Albeit speculative, such processes could include inhibition of locomotion/feeding activity through neuronal effects of Cu (e.g. Untersteiner et al. 2003) or inhibition of digestive enzyme activity (e.g. Chen et al. 2002). It has been shown before that metals can affect Ca uptake (e.g. Brix et al. 2012) and feeding activity (De Schampelaere et al. 2008), but it remains unresolved whether a mechanistic relation between both effects exists. Regardless, it should be clear that our DEB modelling cannot provide any information on the “upstream” cause of reduced assimilation nor on the primary cause of reduced snail growth and reproduction. Detailed time-course studies that include parallel observations of individual level effect of copper (e.g. on growth), feeding activity and assimilation rates, and biomarkers of hypothetical upstream mechanisms (such as those quoted above) are needed to establish the primary cause of toxicity and, eventually, the adverse outcome pathway. Fortunately, we do not need to know all of this to be able to estimate the toxic effects on DEB processes so that “downstream” effects at the individual and population level can be computed. In absence of further mechanistic information, incorporating toxic effects in a DEB-IBM with the best fitting PMoA is currently the best available method to extrapolate toxic effects of Cu for *L. stagnalis* across levels of biological organization (from DEB processes to the individual level, and from the individual to the population level with IBM).

4.3. Food source effect at the PMoA level

The effect of the food source on Cu toxicity to the snail at the individual level is also reflected at the PMoA (Table 4) and population level (Table 5). This could be explained as a consequence

of the separate calibration for lettuce and fish flakes, where it was assumed that the food source interferes with the sensitivity to Cu. Nonetheless, the effects at the PMoA level between both food sources (Figure 5b) are situated closer to each other than the effects on the apical level (Figure 5a). This is most clear in the normalized dose-response curves at the PMoA level (right hand side of Figure 5b). The curves for lettuce and fish flakes are situated close to each other, indicating that the concentration-dependent effect of Cu on the PMoA is similar (i.e. a similar sensitivity between treatments). The residual difference in EC10 values at PMoA level between the lettuce and the fish flakes treatment suggests that food source somewhat affects Cu sensitivity at the PMoA level, but this effect is clearly less important than at the individual level. In addition, a detailed DEB-model analysis of energy fluxes in *Lymnaea* (see Supplemental Information, Section 7) suggests that higher sensitivity to chemicals at individual level at lower food availability or food quality (and thus lower assimilation under control conditions) automatically emerges from DEB principles when the PMoA is inhibition of energy assimilation. Thus, DEB theory seems able to mechanistically explain an important portion of the variability between individual-level effects of copper in *L. stagnalis* when the food source is different.

4.4. Food source effect at the population level

Compared to individual-level toxicity data, the variability in population-level effect concentrations for Cu (EC10 values) to *L. stagnalis* is much smaller (Table 6; Figure 5). Individual-level EC10 values are observed between 1.13 and 78 µg/L (70-fold variation), whereas population level EC10 values were between 2.9 and 44.6 µg/L (15-fold variation). For lettuce, there is an increase in EC10 when extrapolating from the individual to the population level, i.e. from 7 µg/L for the endpoint weight to 14.4 µg/L for the population reproduction. For fish flakes, the opposite was observed, the predicted population EC10 (i.e. 44.6 µg/L for adult

density) is lower compared to the lowest observed EC10 at the individual level (i.e. 78 µg/L for weight). Brix et al. (2011) reported the lowest ECx values for an apical endpoint at the individual level (Table 1). At the population level, we predicted an increase in EC10 for Brix et al. (2011) (i.e. from 1.13 µg/L at the individual level, to 2.9 µg/L at the population level – endpoint: population reproduction), suggesting the population is less sensitive to Cu than the individuals (Table 6 – mixed food type). The range in predicted EC10 values between food sources decreases when going from the apical level to the population level. This trend is best observed in Figure 5 (left side – non-normalized), where the dose-response curves move towards each other when moving from the individual to the population level. The food source effect is thus less important at the population level.

A hypothesis to explain the predictions at the population level in Figure 5 is that the dynamics of populations under stress are governed by a compensation mechanism between two stressors, in this case Cu toxicity effects and food limitation. The simulations were performed assuming a constant food level. Under such conditions the population will reach an equilibrium density when food resources become limiting (Figure 4 – control). Exposure to Cu will decrease energy assimilation in sensitive individuals, which will cause the individuals to be smaller. Smaller individuals will consume less food, leaving more food for others and this leading to other individuals not or less being limited by food. Thus, Cu-induced reduction of energy assimilation is compensated by less strong competition for food. Barata et al. (2002) have already put forward this hypothesis when investigating the dynamics of cladoceran populations. They hypothesized that at food-limiting conditions, the effects of toxicants are compensated by density-dependent effects. Since more food is available for survivors, higher production rates in the population could be reached. A similar trend was observed in the population experiments from Gui et al. (2008) where they describe the joint effects of toxicants and food availability

on population persistence of the fruit fly. They stated that the effect of a stressor may reduce the stress of food limitation and increase resource efficiency. Similarly, Gao et al. (2016) have reported a decrease in Zn toxicity to algae under nutrient-limiting conditions. In conclusion, it is possible that copper toxicity compensates the food limitation effect and that this mechanism evens out the sensitivity of *L. stagnalis* to Cu under different food sources at population level.

4.5. Comparison between studies and individual and population-level endpoints

The variation in EC10 values for dissolved Cu is reduced after extrapolation to the population level (Table 6 – from a 70-fold variation at the individual level, to a 15-fold variation at the population level), but these results do not yet include the effect of water chemistry on Cu bioavailability. Brix et al. (2011) have used a lower DOC content in their test set-up compared to Berteloot et al. (2015), which might explain the higher sensitivity to Cu observed by the authors. A correction with a biotic ligand model (BLM) was performed to consider the effect of bioavailability. The dose-response curves at both the individual and population level were thus corrected for the water chemistry (Table 6; Figure 5). Nonetheless, at the individual level, the EC10 values are still a factor 53 apart, indicating the food source or genetic strain still has a large effect on the snails' sensitivity to Cu. At the population level, the variation in EC10 values is reduced more strongly as the minimum and maximum EC10 values are only a factor 7.4 apart after normalization (Table 6). A portion of the variation reported earlier (a factor 15 at the population level) could thus be appointed to the differences in water chemistry (i.e. bioavailability) between the studies. The remaining difference in sensitivity between the *L. stagnalis* populations from the Berteloot et al. (2015) and Brix et al. (2011) studies could be explained by genetic differences in Cu sensitivity between the European and North-American populations of *L. stagnalis* (Côte et al. 2015) and natural (biological) variation.

It is clear from Figure 5 that the variability on the (normalized) EC10 value is decreased, as the ratio between the maximum and minimum EC10 values is much lower. If we move from the apical level to the PMoA level, and eventually to the population level, we can see that the distance between the minimum and maximum EC10 values decreases (highlighted by the grey areas in Figure 5), indicating a decrease in the variability on effect concentrations. Testing this hypothesis experimentally in the future may help to validate whether our DEB-IBM, calibrated on individual-level chronic effects, is able to accurately predict effects at the population level.

4.6. Modelling-based ecological risk assessment

Although in this exercise the data is lacking to verify the model predictions at population level, new and valuable information can be derived from the reported model development and performed simulations (i.e. the predictions). By integrating toxicity data from different studies with comparable species (i.e. *L. luteola* and *L. stagnalis*), we were able to infer assimilation inhibition as a plausible pMoA for Cu toxicity in *Lymnaea* spp. (excluding the possibility of multiple simultaneous pMoA's). This PMoA explained the observed effect of Cu on growth and reproduction, but is not necessarily the primary cause, and also partly explains the food source effect on copper toxicity. Our modelling thus generated a hypothesis that could be tested with new data and/or experiments. Studies in which in vivo observations of cellular effects are made in parallel with observations of effects on feeding and growth are needed to establish the link, if any, between cellular toxicity of copper and the putative (model-derived) decrease in energy assimilation. Ashauer and Jager (2018) have proposed to link cellular responses to the energy budget (DEB) of an organism as an important avenue for further research in ecotoxicology.

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Extrapolation of effects from the individual to the population is predicted to decrease the uncertainty on effects of Cu on *L. stagnalis*. Different food sources lead to divergent effect concentrations at the apical (individual) level. The effect of the food source is still present at the population level, albeit less prominent. A potential way forward would be to extrapolate different food sources and effects from different genetic strains to the population level and derive one general threshold for *L. stagnalis*. Moreover, it is worthwhile to do this for other species as well, and as such a species sensitivity distribution could be established at the population level. The derived environmental threshold concentration would then account for a shared (common) population-level effect, instead of the mix of apical endpoints in current ERA. Hence, with this study, we want to promote the use of population modelling in risk assessment of metals, and by extension other chemical stressors.

5 CONCLUSION

With some adaptations to an existing DEB-IBM (i.e. metabolic acceleration, the assimilation correction factor and Cu toxicity), we extrapolated chronic Cu toxicity effects for different studies (different food sources and genetic strains) from the individual to the population level for *L. stagnalis*. We explained the variability in sensitivity observed between food sources (lettuce vs fish flakes) with a DEB model for *L. stagnalis*. Calibration of our model suggest that inhibition of energy assimilation is the most plausible physiological mode of action (PMoA) that can explain the effect of the food source on chronic Cu toxicity in *Lymnaea* spp. Our study shows the importance of integrating data on different endpoints (simultaneous growth and reproduction) from related species (i.e. same genus) to identify the possible PMoA of a chemical stressor. Toxicity effects observed at the individual level were translated with DEB-IBM to the population level, although data were lacking to validate our model at the population level. Based on a series of simulations, the food source effect is predicted to be less important at the population level. The EC10 values on population equilibrium density for lettuce and fish flakes were only a factor 2.3 apart (based on dissolved concentrations) compared to a factor 11 at the individual level. Comparing individual-level to population-level endpoints from different studies, EC10 values are situated in a smaller concentration range. The range in EC10 values decreased from a factor 70 at the individual level to a factor 15 at the population level. With a bioavailability correction to account for differences in-water chemistry this was even further reduced to a factor 7.4. Extrapolation to the population level reduced variability among threshold levels from different studies. We therefore encourage a transition to a more modelling-based environmental risk assessment.

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SUPPLEMENTAL INFORMATION

1. **Figures S1-S7.**
2. **Tables S1-S3.**
3. **Figshare: raw data from Berteloot et al. (2015).**
URL <https://figshare.com/s/7ae21849c55877c9a762>
4. **Figshare: zip archive with the Lymnaea DEB-IBM.**
URL <https://figshare.com/s/b628fd3165636ec69a80>
5. **Pseudo code for the DEB-IBM model**
6. **Details on the Biotic Ligand Correction**
7. **Additional DEB analysis on an energetic level**

ACKNOWLEDGEMENT

This project is supported by Copper Alliance (European Copper Institute). K. Vlaeminck is the recipient of the PhD grant *Baekeland-mandaat* provided by VLAIO (Vlaams Agentschap voor Innoveren en Ondernemen) and supported by ARCHE (Assessing Risks of Chemicals) Consulting.

FIGURE CAPTIONS

Figure 1. Predicted (lines) and observed (dots) snail size over time, for both lettuce (a) and fish flakes (b) (Berteloot et al., 2015). The blue dots represent the observed snail size (*L. stagnalis*). The green squares are the mean of each assessed time point. The black lines indicate the minimum, maximum and average prediction without an assimilation correction factor. The red lines indicate the minimum, maximum and average prediction with the calibrated assimilation correction factor.

Figure 2. Observed and predicted reproduction (left) and growth (right) as a function of the copper concentration (Das and Khangarot 2011). Data from Das and Khangarot (2011) (green squares) and model predictions (grey area) after 7 weeks of exposure to copper are plotted. The lines represent the maximum, minimum and mean prediction. Both growth and reproduction are represented relative to the control. Results are shown for the PMoA for assimilation (a), the PMoA for cost of growth (b), and the PMoA for maintenance (c). No assimilation correction factor was considered, and effects were calibrated relative to the control.

Figure 3. Predicted snail growth as a function of the dissolved copper concentration for the PMoA assimilation (i.e. a decrease in energy assimilation with increasing copper concentrations), for both lettuce (a) and fish flakes (b). Data on snail size (blue dots) are plotted against the model predictions (black lines). The green squares represent the mean observed size of each concentration. The lines represent the maximum, minimum and mean predicted size. Data on day 28 of the experiment of Berteloot et al. (2015) is shown. The respective value for the assimilation correction factor for lettuce ($a_f = 1.3$) or fish flakes ($a_f = 4.0$) was used.

Figure 4. Example of a simulated *L. stagnalis* population. Population biomass over time is shown, assuming constant environmental conditions (i.e. constant food level, water chemistry, temperature and exposure concentration), for lettuce (a) and fish flakes (b) (with their respective assimilation correction factor and effect parameters). The grey area indicates the control simulations (i.e. no copper exposure). The red area shows the population when exposed to 30 $\mu\text{g Cu L}^{-1}$.

Figure 5. Comparison between the dose-response curves on apical (a), PMoA (b) and population level (c), based on extrapolation of effects observed by Berteloot et al. (2015) (lettuce – blue curve, and fish flakes – red curve) and Brix et al. (2011) (green curve). The left figures give the dose-response curves in dissolved Cu concentration. The right figures show the same dose-response curves, after normalization, in Cu concentration on the biotic ligand (BL). The dashed vertical lines represent the EC10 values of the respective dose-response curve.

Table 1. Overview of the literature concerning copper toxicity on the pulmonate snails, *L. stagnalis* (the great pond snail) and *L. luteola* (the apple snail). The reported effect concentrations (ECx values as µg dissolved Cu/L) and the initial life stage are presented, along with the exposure duration of the experiment. The reported ECx are shown. – Note that for Khangarot and Das (2010) and Das and Khangarot (2011) no info is given on some ECx values, although data is available on these endpoints. The 95% confidence intervals are given between parenthesis (only for studies where the confidence intervals were reported).

Reference	Species	Effect of copper	Initial life stage	Exposure duration	ECx	Value (µg/L)
Khangarot and Das (2010)	<i>L. luteola</i>	Embryo survival	Embryo	96 h	LC50	28.3 (21.9 – 36.7)
		Embryo survival	Embryo	240 h	LC50	18.2 (16.7 – 19.3)
		Embryo survival	Embryo	336 h	LC50	15.9 (15.0 – 17.2)
		Embryonic development	Embryo	14 d	-	-
		Hatching success	Embryo	10 d	EC50	2.2 (2.0 – 2.6)
Brix et al. (2011)	<i>L. stagnalis</i>	Survival	Juvenile	96 h	EC50	30.7 (28.9 – 32.7)
		Wet weight	Juvenile	30 d	EC20	1.8 (1.3 – 6.1)
		Wet weight	Juvenile	30 d	EC10	1.13*
		Survival	Juvenile	30 d	LC20	5.6 (4.8 – 6.7)
Das and Khangarot (2011)	<i>L. luteola</i>	Survival	Adult	7 d	LC50	63.6 (56.9 – 70.2)
		Survival	Adult	14 d	LC50	40.9 (37.2 – 44.8)
		Survival	Adult	21 d	LC50	20.9 (17.1 – 25.8)
		Survival	Adult	28 d	LC50	15.9 (13.2 – 18.7)
		Survival	Adult	49 d	LC50	9.5 (8.42 – 10.7)
		Growth rate	Adult	49 d	-	-
		Number of eggs	Adult	49 d	-	-
		Food consumption	Adult	49 d	-	-
		Embryo survival	Embryo	49 d	-	-
		Egg mass	Embryo	49 d	-	-
		Hatching success	Embryo	49 d	-	-
Ng et al. (2012)	<i>L. stagnalis</i>	Survival	Juvenile	96 h	LC50	24.9 (19.6 – 30.4)
		Survival	Juvenile	96 h	LC20	18.0 (12.1 – 22.3)
Das et al. (2012)	<i>L. luteola</i>	Survival	Adult	96 h	LC50	92.7 (75 – 112)
Berteloot et al. (2015)	<i>L. stagnalis</i>	Survival (lettuce)	Juvenile	28 d	LC50	128.0 (113 – 145)
		Growth rate (lettuce) – length	Juvenile	28 d	EC50	98.0 (77 – 119)
		Growth rate (lettuce) – length	Juvenile	28 d	EC20	34.0 (20 – 48)
		Growth rate (lettuce) – length	Juvenile	28 d	EC10	17.0 (6 – 28)
		Dry blotted weight (lettuce)	Juvenile	28 d	EC50	51.0 (27 – 76)
		Dry blotted weight (lettuce)	Juvenile	28 d	EC20	15.0 (3 – 27)
		Dry blotted weight (lettuce)	Juvenile	28 d	EC10	7.0 (-1 – 16)
		Survival (fish flakes)	Juvenile	28 d	LC50	109.0 (73 – 163)
		Growth rate (fish flakes) – length	Juvenile	28 d	EC50	110.0 (-1379 – 1616)
		Growth rate (fish flakes) – length	Juvenile	28 d	EC20	90.0 (77 – 103)
		Growth rate (fish flakes) – length	Juvenile	28 d	EC10	79.0 (32 – 136)
		Dry blotted weight (fish flakes)	Juvenile	28 d	EC50	92.0 (15 – 170)
		Dry blotted weight (fish flakes)	Juvenile	28 d	EC20	84.0 (-83 – 251)
		Dry blotted weight (fish flakes)	Juvenile	28 d	EC10	78.0 (-138 – 296)
Côte et al. (2015)	<i>L. stagnalis</i>	Survival	Juvenile	96 h	LC50	53.0

*Value was recalculated from the dose-response data from the original publication, using a three-parameter log-logistic model (drc package in RStudio; Ritz et al., 2015).

Table 2. DEB parameters for *L. stagnalis* as reported by Zimmer et al. (2014).

Parameter	Symbol	Value	Unit
Maximum surface-specific assimilation rate	p_{Am}	25.2049	J/(d*cm ²)
Maximum surface-specific searching rate	F_m	6.5	L/(d*cm ²)
Energy conductance	ν	0.01394	cm/d
Allocation fraction to the soma (structural growth and structural maintenance)	κ	0.6876	-
Reproduction efficiency	κ_R	0.475	-
Volume-specific somatic maintenance rate	p_M	58.57	J/(d*cm ²)
Specific costs for structure	E_G	3900	J/cm ³
Maturity maintenance rate	k_J	0.002	d ⁻¹
Maturity at birth	E_H^b	0.285	J
Maturity at transformation	E_H^j	50.1	J
Maturity at puberty	E_H^p	496.6	J
Weibull ageing acceleration	h_a	1.287E-5	d ⁻²
Gompertz stress coefficient	s_G	0.0001	-
Arrhenius temperature	T_A	8000	K

Table 3. DEB parameters under stress, taken from Martin et al. (2013b). The mathematical equations of the stressed value differ however from the implementation from Martin et al. (2013b). They used a hockey stick relationship for the stress level, whereas here a log-logistic relationship was chosen.

PMoA	Description	Affected parameter	Stressed value
Feeding/assimilation	Decrease in energy assimilation efficiency	f	$f_s = f * (1 - s)$
Maintenance	Increase in maintenance costs	k_M	$k_{M,s} = k_M / (1 - s)$
		k_J	$k_{J,s} = k_J / (1 - s)$
Growth costs	Increase in overhead costs of growth	k_M	$k_{M,s} = k_M * (1 - s)$
		g	$g_s = g / (1 - s)$
Reproduction costs	Increase in cost per egg	k_R	$k_{R,s} = k_R * (1 - s)$
Embryonic hazard	Decrease in survival during embryonic period	k_R	$k_{R,s} = k_R * \exp(-s)$

Table 4. Results of the calibration of the effect parameters for the dataset of Berteloot et al. (2015). Results are based on the 100 best fits (based on the Euclidean distance ρ) of 10,000 simulation runs. Model predictions were fitted against experimental data on snail size from the dataset of Berteloot et al. (2015) for both fish flakes and lettuce. Calibration for the food sources (lettuce and fish flakes) was done separately.

PMoA	Food type	$EC50_{PMoA}$ ($\mu\text{g/L}$)		β_{PMoA} (-)		ρ
		avg	sd	avg	sd	
Inhibition of assimilation	Lettuce	64.1	0.3	1.67	0.02	1.09
	Fish flakes	77.4	0.2	4.70	0.02	1.18
Increase in costs for growth	Lettuce	48.6	0.2	2.19	0.02	2.94
	Fish Flakes	77.4	0.2	4.61	0.03	5.49
Increase in cost for maintenance	Lettuce	62.4	0.4	4.01	0.05	3.51
	Fish Flakes	77.2	0.2	4.51	0.05	9.76

Table 5. Results of the population simulations for the effect parameters of Berteloot et al. (2015) (lettuce and fish flakes) and Brix et al. (2011) (mixed food). The predicted ECx values for every population-level endpoint is shown, with the corresponding standard error. Results based on 100 iterations. A comparison is made between the two food types: lettuce ($a_f = 1.3$), fish flakes ($a_f = 4.0$) and mixed food ($a_f = 3.8$).

Population variable	Unit	Food type	EC10 ($\mu\text{g/L}$)		EC20 ($\mu\text{g/L}$)		EC50 ($\mu\text{g/L}$)	
			avg	se	avg	se	avg	se
Equilibrium population density	#/ m^2	Lettuce	15.6	0.7	23.5	0.3	47.1	1.1
		Fish flakes	47.8	2.6	53.4	2.1	65.5	1.3
		Mixed	3.1	0.2	4.0	0.2	6.4	0.1
Embryo density	#eggs/ m^2	Lettuce	16.2	0.6	24.7	0.4	51.0	1.8
		Fish flakes	50.3	3.3	55.5	2.6	65.8	1.6
		Mixed	3.0	0.3	3.9	0.3	6.1	0.2
Juvenile density	#/ m^2	Lettuce	15.7	0.7	23.3	0.5	46.1	1.7
		Fish flakes	49.0	3.3	54.4	2.7	65.0	1.6
		Mixed	3.1	0.3	4.0	0.3	6.1	0.2
Adult density	#/ m^2	Lettuce	15.1	0.4	21.5	0.3	39.5	0.6
		Fish flakes	44.6	4.0	50.5	3.3	62.4	2.0
		Mixed	3.6	0.4	4.4	0.4	6.3	0.2
Population reproduction	#eggs per adult per day	Lettuce	14.4	0.5	21.4	0.4	42.1	0.8
		Fish flakes	47.2	2.5	53.1	2.0	64.9	1.3
		Mixed	2.9	0.2	3.8	0.2	6.0	0.2

Table 6. Individual-level and population-level EC10 values for *L. stagnalis*. The values are expressed as dissolved Cu concentrations and, after normalization for bioavailability, as copper concentrations on the biotic ligand (BL). Only values for the most sensitive individual and population-level endpoints are given. Individual-level endpoints: dry blotted weight for Berteloot et al. (2015); growth rate (wet weight) for Brix et al. (2011). Population-level endpoints: population reproduction for the lettuce treatment from Berteloot et al. (2015) and Brix et al. (2011); adult density for the fish flakes treatment from Berteloot et al. (2015).

Reference	Food type	EC10 _{ind}		EC10 _{pop}	
		Dissolved (µg Cu/L)	Normalised (nmol/gw BL)	Dissolved (µg Cu/L)	Normalised (nmol/gw BL)
Berteloot et al. (2015)	Lettuce	7.0	1.68	14.4	2.76
	Fish flakes	78.0	9.54	44.6	6.25
Brix et al. (2011)	Mixed	1.13	0.18	2.9	0.85

Table 7. Dry matter content (DMC) and protein content of various food sources for *L. stagnalis*.

Food source	DMC (% of wet weight)	Protein content (% of DMC)	Protein content (% of wet weight)	Reference
Organic lettuce	6.6%	6.9%	0.46%	Berteloot et al. (2015)
Macrophyte	5 – 23%	6 – 13%	0.3 – 2.99%	Elger and Lemoine (2005)
Biofilm	3 – 10%	5 – 20%	0.15 – 2%	Fernandes da Silva et al. (2008)
Macrophyte detritus	18 – 22%	3 – 24%	0.45 – 5.28%	Rice (1982)
Periphyton	10%	13 – 32%	1.3 – 3.2%	Sladeczek and Sladeczkova (1964)
Lettuce	5 – 12%	20 – 40%	1 – 4.8%	McKeehen et al. (1996); Selck et al. (2006)
Fish flakes (TetraMin)	94%	46%	43.2%	Tetra GmbH

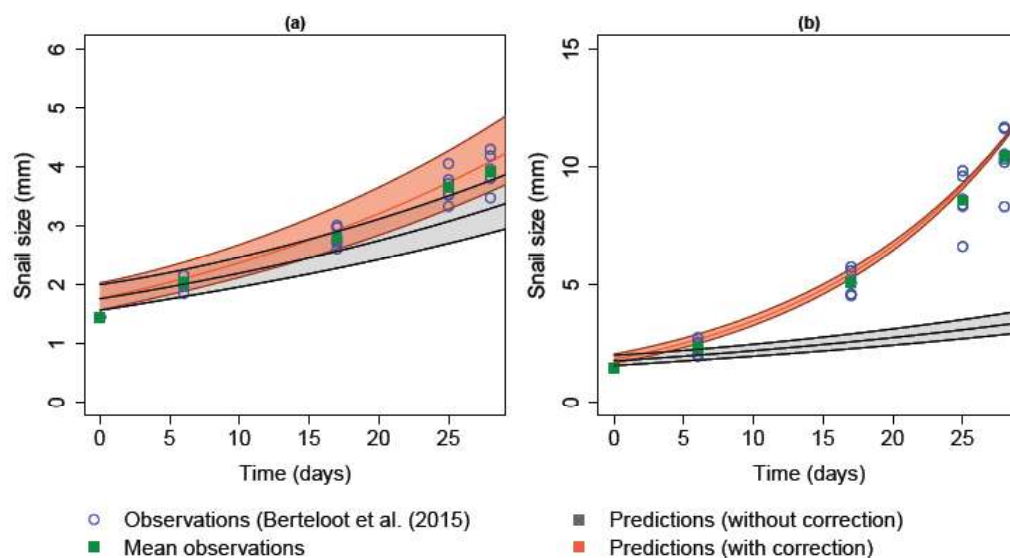


Figure 1

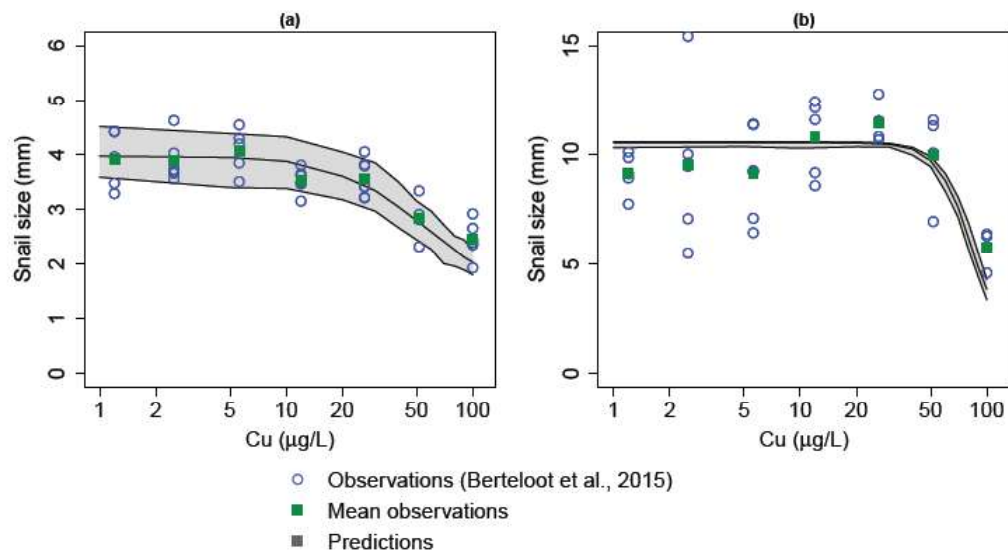


Figure 2

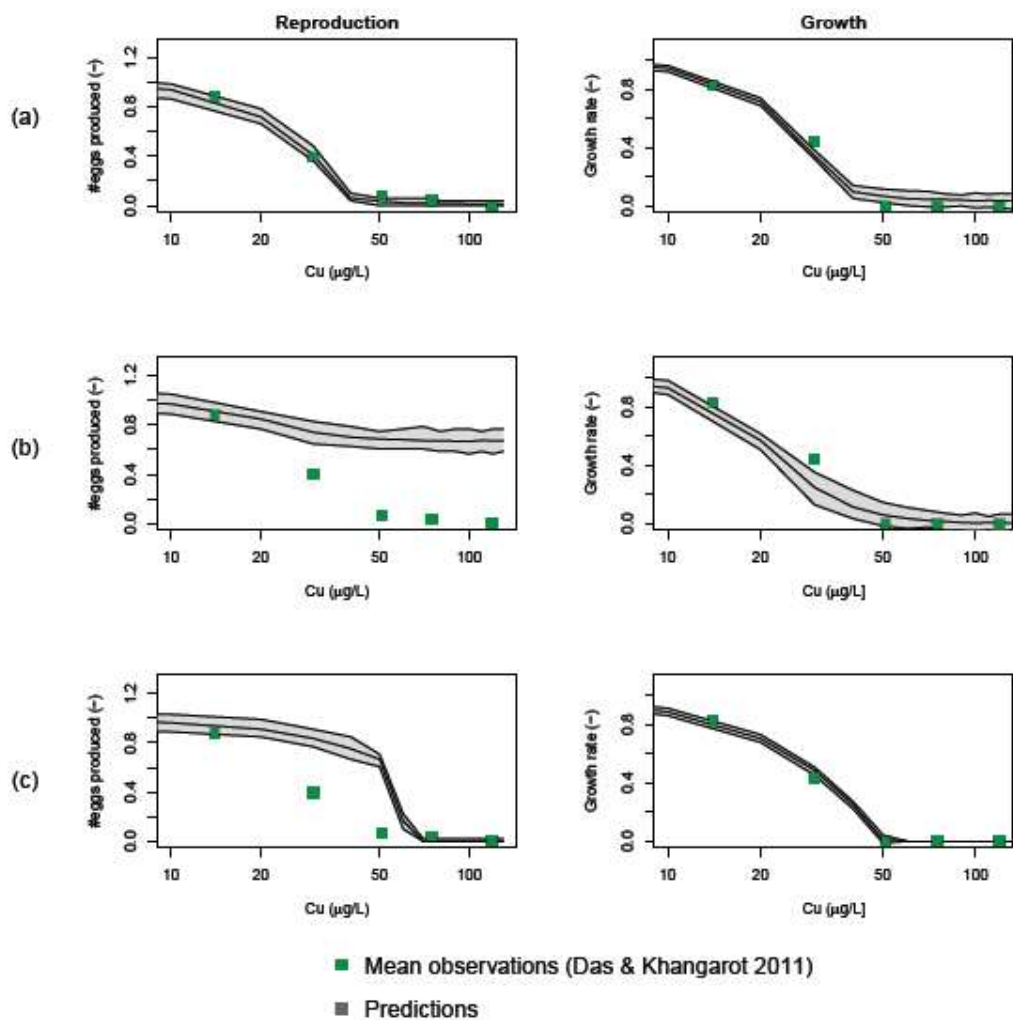


Figure 3

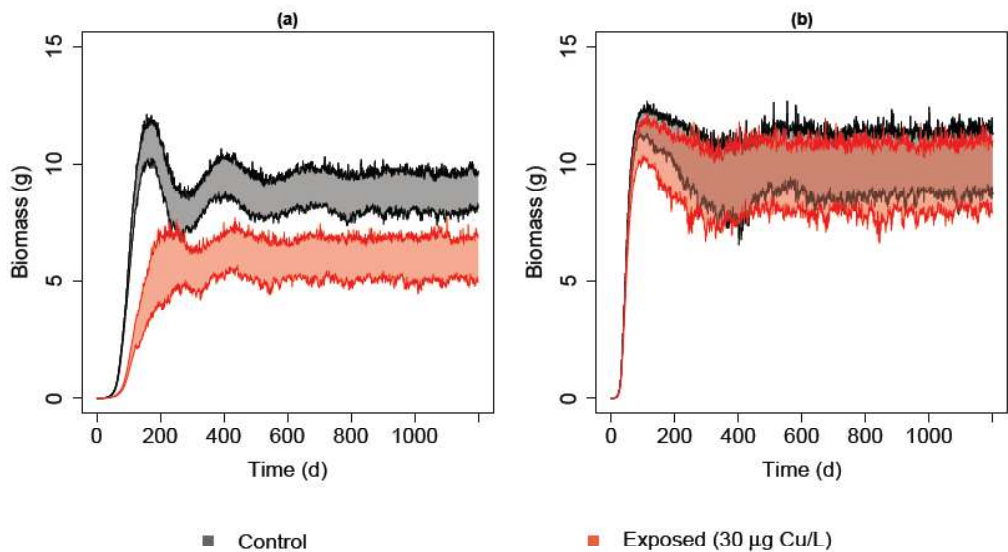


Figure 4

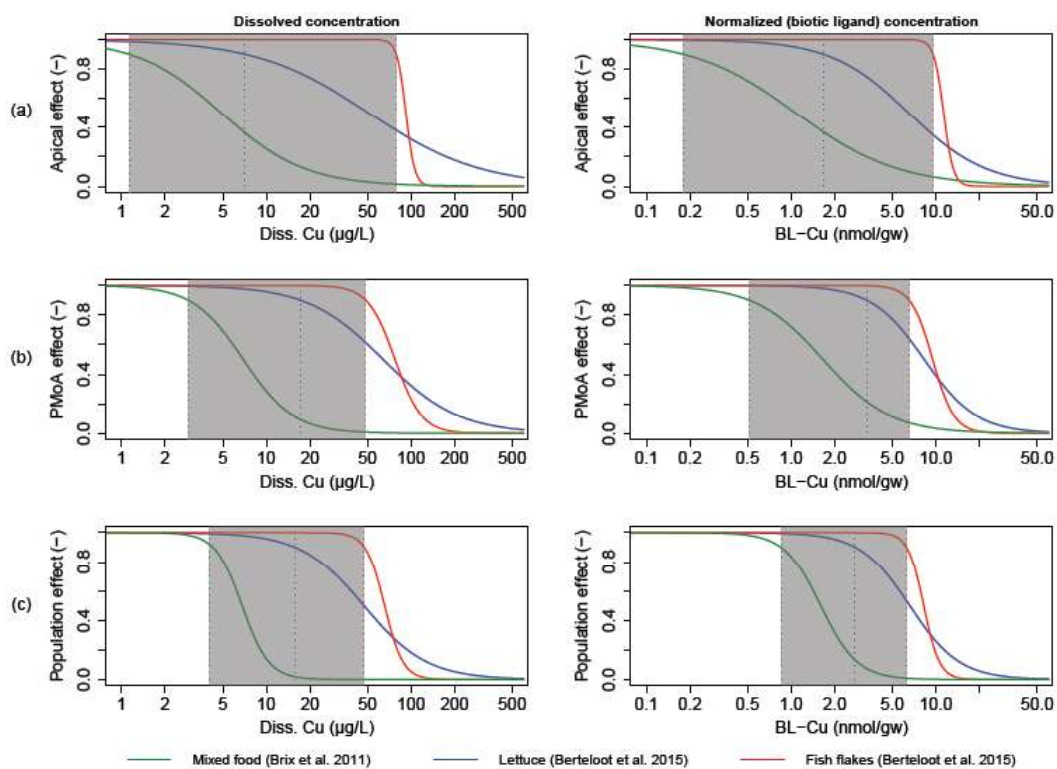


Figure 5